

Sensitivity Study of the Wigner-Smith Operator: Applications in Electromagnetism

Mohamed Amine Ezzine ✉

Department of Electronics, Optronics and Signal Processing
Institut Supérieur de l'Aéronautique et de L'espace

Antoine Saucourt ✉

Department of Electronics, Optronics and Signal Processing
Institut Supérieur de l'Aéronautique et de L'espace

Valentin Mazieres ✉

Department of Electronics, Optronics and Signal Processing
Institut Supérieur de l'Aéronautique et de L'espace

Abstract

The Wigner-Smith (WS) operator, in its generalized form, serves as a powerful tool for identifying optimal wave shapes for specific tasks. These tasks can include minimizing dispersion while traversing a medium or focusing energy at a desired point.

This operator relies on the derivative of the scattering matrix with respect to a variable of the interest (slab position, refractive index of a region, etc.). Consequently, when using this operator in practice, interacting with the system could lead to other unwanted changes.

This work proposes a simplified framework to investigate the impact of such unwanted changes on the behavior of the WS operator. The system under study is a 4-port system traversing a scattering region with a slab. First the eigenstates associated with a derivation with respect to the frequency and then the slab position are computed and obtained in simulation. Finally, we focus on the study of the Wigner-Smith operator when both changes are present. We demonstrate that the resulting eigenstates cannot be described as simple linear combinations of states arising from individual variations. Instead, the combined effect depends on the relative importance of the variations. This dependence can manifest as either a dominant influence from one variation or a more complex interplay between them.

Keywords

Generalized Wigner Smith Operator, Wavefront Shaping, Sensitivity Study, Multi-variable Variation

Introduction

Electromagnetic waves propagating through complex scattering systems undergo multiple

diffraction phenomena. This results in intricate energy distribution patterns that can initially appear entirely random. In most cases, only a minor portion of the wave energy successfully transmits through the system. This poses a significant challenge for communication applications [1]. Furthermore, the presence of disorder within complex systems makes it difficult to control the behavior of the waves within them. Focusing on a target within such a system is an essential challenge, for micro body manipulation for example [2].

In the study of scattering phenomena, the scattering matrix, often denoted by S , serves as a valuable tool. This matrix relates the outgoing wave (output) to the incoming wave (input) after interacting with the system, as described by the following equation:

$$\Phi_{out} = S\Phi_{in} \quad (1)$$

The scattering matrix plays a pivotal role in characterizing the physical properties of a scattering system. Understanding its behavior offers valuable insights into manipulating and controlling scattering phenomena.

One intriguing feature of the scattering matrix emerges when multiple propagation channels interact. Contrary to the expectation of uniform energy distribution across these channels, they often exhibit a tendency to segregate into nearly-open or nearly-closed channels. This phenomenon was first theoretically predicted using Random Matrix Theory (RMT) [3]. Subsequently, John Pendry et al. [1] conducted an experiment demonstrating the excitation of these open channels. They achieved this by manipu-

lating the incident angle and phases of light arrays impinging on a scattering system composed of disordered zinc oxide particles. This manipulation led to a remarkable transmission enhancement of a factor of 10^3 .

Another exciting development emerged from the field of microbody manipulation. In 2003, Shoji Maruo et al. [4] demonstrated the use of electromagnetic waves to control micromanipulators and perform specific tasks. However, their approach required a high-power laser to effectively focus the energy onto the micromanipulator.

These experimental results showcase the potential to overcome the apparent randomness within scattering systems. This paves the way for a crucial question: how can we determine the optimal wave shape to achieve specific goals within a scattering system? This could involve exciting open or closed propagation channels, focusing energy on a target micromanipulator, or various other objectives.

The Wigner-Smith (WS) and Generalized Wigner-Smith (GWS) operators offer valuable tools for analyzing wave propagation phenomena. Depending on the variable with respect to which the operator is derived, its eigenstates correspond to wave shapes that extremize specific properties [5].

In this work, we focus on the classical WS operator derived for frequency variation and the GWS operator derived for variation in scatterer position. We aim to reproduce established results for both cases before investigating the combined effect of multi-variable changes. Our analysis is conducted within the framework of a simplified four-port system, providing a clear platform for understanding the operators' properties.

This article is divided into three parts.

1. Theoretical background and model presentation
2. Visualization of the eigenstates of the WS and GWS operator on our system
3. Sensitivity analysis of the combined effect of frequency and slab position variations on the GWS Operator

1. Theoretical background and model presentation

1.1. Wigner-Smith operator

1.1.1. Introduction to the Wigner-Smith operator

Originally, Wigner introduced the WS operator [2] in the field of quantum physics to calculate the delay time between the absorption and emissions of photons during a scattering event [6]. Later Smith extended its use in multichannel scattering systems [7]. (See appendix 4.2 for a theoretical derivation of The Wigner Smith operator).

$$Q = -iS^\dagger \frac{\partial S}{\partial f} \quad (2)$$

Instead of a simple analysis of the scattering matrix, the WS operator offers an elegant tool to study the behavior of waves inside a complex medium.

For a linear, time-invariant reciprocal system, the scattering matrix S is unitary, i.e. it satisfies equation 3 [8].

$$S^\dagger S = I \quad (3)$$

For such a system the corresponding WS operator is hermitian, i.e it satisfies the equation 4.

$$Q^\dagger = Q \quad (4)$$

It follows from this that, for a given incident field Φ_{in} of size n , the expected value of the operator, given by equation 5, is always a real number that could be interpreted as the average time the wave spends inside the system [8].

$$\langle \Phi_{in} | Q | \Phi_{in} \rangle = \sum_{i=1}^n |S\Phi_{in}|_i^2 \frac{\partial \phi_i}{\partial \omega} \quad (5)$$

where ϕ_i is the phase of $(S\Phi_{in})_i$ and ω is the angular frequency.

Building on this concept and using the Courant Fisher minmax theorem, one can prove that the eigenvalues of the WS operator extremize the time a wave spends inside a system [8]. Indeed, the different eigenvectors of the WS operator correspond to wave functions that exhibit particle-like behavior. Instead of following a complex pattern of propagation inside the system, these states bounce on the edges as if they were classical particles following a well defined trajectory [9]. Respectively, The eigenvector with the smallest (largest) associated eigenvalue correspond to a wave that takes the shortest (longest) path inside a scattering medium [8].

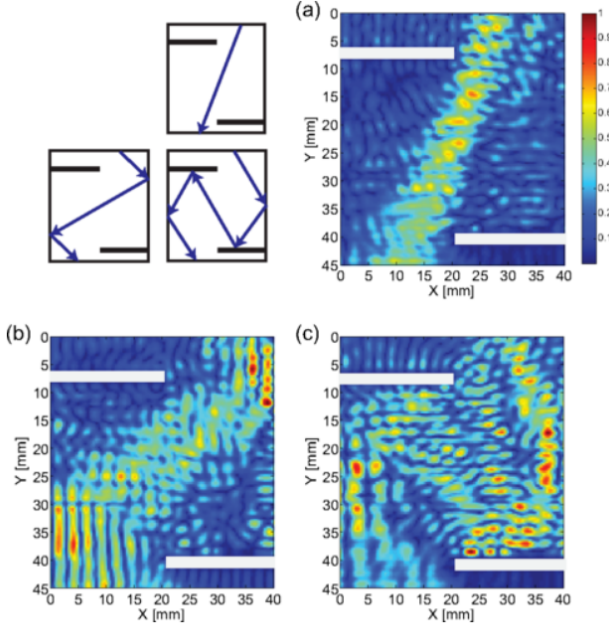


Figure 1: three particles like states produced by Gérardin et al [10]. The wave function exhibits a clear trajectory within the scatterer, deviating from the expected complex diffraction and dispersion patterns.

In 2016, Gérardin et al. used the eigenstates of the WS operator to produce wave forms that exhibit this particle-like state illustrated in Figure 1 [10].

The WS operator has proven to be a powerful tool for determining the specific wave shape to excite the open or closed channels of a scattering system.

1.1.2. A simplified 2-port system : an analytical characterization of the expected value of the WS operator

In this section we focus on a simplified system, figure 2. The goal is to analytically calculate the expected value of the WS operator.

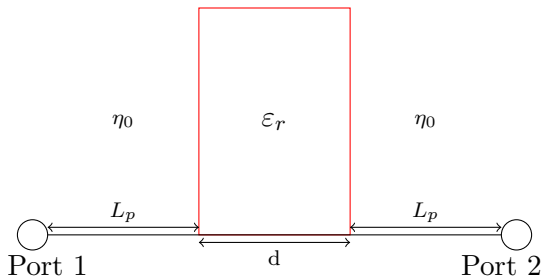


Figure 2: 2 port system : The slab's width is d while the 2 ports are place symmetrically on both sides of the slab with a distance L_p

The scattering matrix that describes this geometry is given by :

$$S = \begin{Bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{Bmatrix} \quad (6)$$

Where S_{ij} is the scattering coefficient measured in port j due to excitation in port i .

The different coefficients of S matrix could be expressed as a function of the reflection and transmission coefficient of the Slab Γ and τ respectively.

1. $S_{11} = S_{22} = \Gamma$
2. $S_{12} = S_{21} = \tau$

Where Γ and τ are given by [11]

$$\Gamma = \frac{\rho(1 - e^{-i\omega T})}{1 - \rho^2 e^{-i\omega T}} \text{ and } \tau = \frac{(1 - \rho^2)e^{-i\omega \frac{T}{2}}}{1 - \rho^2 e^{-i\omega T}} \quad (7)$$

where $\rho = \frac{\eta_s - \eta_0}{\eta_s + \eta_0}$, the interface reflection coefficient, and $T = \frac{2\eta_s d}{C_0}$, the time it takes the wave to travel back and forth through the slab.

Assuming a single excitation at port 1 by a normalized Φ_{in} , the calculation of the WS operator can be simplified by focusing solely on its first element, Q_{11} . This is because only Q_{11} contributes to the expected value when the excitation originates from port 1.

$$Q_{11} = -i(\Gamma^* \frac{\partial \Gamma}{\partial \omega} + \tau^* \frac{\partial \tau}{\partial \omega}) \quad (8)$$

finally for an unique excitation Φ_{in} at Port 1, the expected value is given by

$$\langle \Phi_{in} | Q \Phi_{in} \rangle = \frac{T}{2} \frac{(1 - \rho^4)}{1 + \rho^4 - 2\rho^2 \cos(\omega T)} \quad (9)$$

Equaion 9 shows that the expected values of the WS operator is proportional to $\frac{T}{2}$, the time it takes the wave to travel through the slab. The proportionality represents the amount of energy that crosses the system. The extremes of the expected value are given by :

$$\langle \Phi_{in} | Q \Phi_{in} \rangle_{min} = \frac{T}{2} \frac{(1 - \rho^2)}{(1 + \rho^2)} \text{ for } \omega = m\omega_0 \quad (10)$$

$$\langle \Phi_{in} | Q \Phi_{in} \rangle_{max} = \frac{T}{2} \frac{(1 + \rho^2)}{(1 - \rho^2)} \text{ for } \omega = \frac{m\omega_0}{2} \quad (11)$$

where $\omega_0 = \frac{2\pi}{T}$ and m an arbitrary positive integer.

If we consider a unitary Φ_{in} , and following the development as in [8], we conclude that

$\langle \Phi_{in} | Q \Phi_{in} \rangle_{max}$ corresponds to the classical trajectory where the wave bounces on the slab. Here $\langle \Phi_{in} | Q \Phi_{in} \rangle_{min}$ corresponds to a classical trajectory in which the waves cross the slab. This coherent, with the results we found, indeed for $\omega = m\omega_0$ the reflection coefficient is null, all the energy passes from port 1 to port 2. While for $\omega = \frac{m\omega_0}{2}$ the reflection coefficient is maximized, most of the wave energy bounces on the slab[11].

In the context of a simple 2-port system, our analysis reinforces our understanding of the WS operator's eigenvalues. However, because there's only a single propagation channel in this scenario, the concept of a shortest propagation path loses its relevance. To explore the WS operator's behavior more comprehensively, we consider a 4-port system. This allows the wave to follow different paths during its propagation from one end to the other.

1.2. The Generalized Wigner-Smith (GWS) operator

1.2.1. Introduction to the generalized Wigner-Smith operator

Although the WS operator was originally derived for frequency variation, a generalized form was recently introduced [5]: Assuming that the scattering properties of a system depends on a parameter α one can define a generalised Wigner-Smith operator (GWS) given by

$$Q_\alpha = -iS^\dagger \frac{\partial S}{\partial \alpha} \quad (12)$$

In 2017, Brandstötter et al [5], showed that if one considers a change in the position of a specific scatterer inside a system, the eigenstates of the GWS corresponds to wave functions that avoid or focus on the scatterer. Furthermore, they showed that the expected value of the GWS operator measures the shift in the conjugate variable to α that the wave experiences when traveling through the scattering medium 13.

$$\langle \Phi_{in} | Q_\alpha | \Phi_{in} \rangle = \langle \Phi_{out} | C_\alpha | \Phi_{out} \rangle - \langle \Phi_{in} | C_\alpha | \Phi_{in} \rangle \quad (13)$$

where $C_\alpha = \frac{-i\partial}{\partial \alpha}$, is the conjugate operator for the global variable α . Table 1 gives some corresponding conjugate variables for a given global variables.

In 2020, based on this work, Horodyski et al [2] utilized the eigenstates of the GWS operator to optimize the applied pressure, torque, or to focus energy on a specific target, regardless of its

Global variable α	Conjugate variable
frequency	dwell time
scatterer position	momentum
scatter angle	angular momentum
Scatterer radius	pressure
refractive index	intensity

Table 1: Corresponding conjugate variables for some given global variables

shape and position, resulting in major advancement in the field of micro object manipulation.

1.2.2. Common WS Operator Computation Scheme and Limitations

When trying to calculate the GWS modes, a common approach involves using the finite difference method to approximate the derivative of the scattering matrix (S matrix) with respect to a global variable α 14.

$$\frac{\partial S}{\partial \alpha} \approx \frac{S(\alpha + \delta\alpha) - S(\alpha)}{\delta\alpha} \quad (14)$$

While this method offers a straightforward approach to evaluate the derivative, it presents a potential limitation. Introducing a small variation ($\delta\alpha$) into the system can inadvertently induce unintended changes to another global variable (β). This unintended coupling can lead to inaccurate calculations of the derivative.

This work aims to characterize the influence of intentional variations on the behavior of (GWS) operator. In particular, we introduce a simplified framework using a 4-port system with a dielectric slab characterized by its width (d) and relative permittivity (ϵ_r). We consider frequency (f) as the primary variable and the position of the slab as the intentional variation. We then examine how different variations in the position of the slab impact the eigenvalues of the main operator, Q_f .

1.3. Model presentation and simulation protocol

1.3.1. Model presentation

The analysis conducted in this work is based on the system presented by figure 3: this four-port system comprises two primary propagation channels connected by a secondary channel. This secondary channel allows the wave to travel between both main channels, adding a necessary layer of complexity for a deeper understanding of the WS operator. The key component of the system is the slab (red in figure 3). This component allows the scattering behavior of the system to be

frequency-dependent while remaining sufficiently simple for straightforward interpretations. Additionally, it provides the system with more flexibility, enabling the study of the GWS operator for various types of variations (position, width, relative permittivity, etc.).

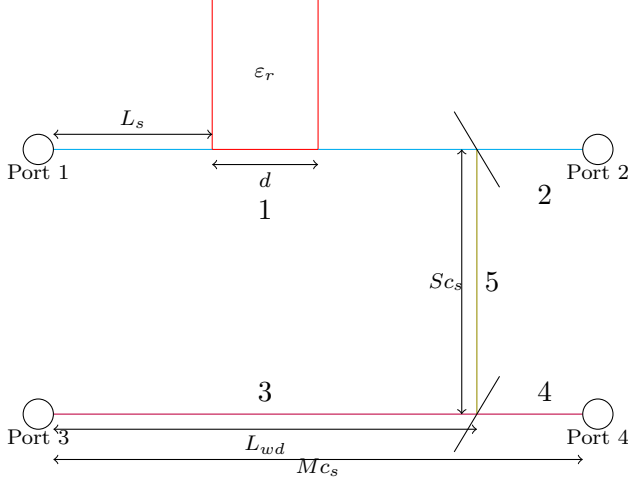


Figure 3: 4-ports system, In blue the first main propagation channel, in purple the second main propagation channel, in green the secondary propagation channel. The two main propagation channels are linked with a secondary channel via 2 wave dividers (in black). These simulation objects represent nodes where two channels meet, at the intersection the wave dividers distribute an incoming wave into the other two channels, by multiplying one side by a coefficient s and the other by a coefficient m . The coefficients are chosen such that $\sqrt{m^2 + s^2} = 1$

Table 2, resumes the different dimensions of the system (here we express the different dimensions in terms of fraction of the main channel size (Mc_s)).

Parameter	Name	Value
Mc_s	main channel size	1
L_s	slab starting position	0.3
d	slab's width	0.2
Sc_s	secondary channel size	0.5
L_{wd}	wave divider position	0.8

Table 2: 4-ports systems, dimension definition

In order to ensure the reciprocity of the system, we fix the values of s and m parameters for each wave divider to :

$$m = s = \frac{1}{\sqrt{2}} \quad (15)$$

Although our 1-dimensional calculation scheme may not fully capture the intricacies of

the real system. Notably, our current definition of wave dividers does not account for wave reflections at the wave dividers. Despite these limitations, this model provides a simplified framework, where we can study the behavior of the WS operator.

1.3.2. Simulation protocol

To simulate the system behavior, we use a one dimensional **Finite-Difference Time-Domain**(FDTD) code. For this we divide the system in 5 different regions.

1. $MC1_{bwd}$: main channel 1 before the wave divider.
2. $MC1_{awd}$: main channel 1 after the wave divider.
3. $MC2_{bwd}$: main channel 2 before the wave divider.
4. $MC2_{awd}$: main channel 2 after the wave divider.
5. SC : secondary propagation channel.

We solve the one-dimensional FDTD equation on each of the domains by introducing the proper source at the start of each of them.

Although theoretically the eigenvalues and eigenstates of the WS operator provide valuable insights, in practice, computational limitations often hinder their accurate calculation. Singular value decomposition (SVD) offers a broader framework for analyzing the WS operator. SVD decomposes a matrix into its singular values and singular vectors, providing information about the "strength" and direction of the operator's action on input vectors. Importantly, SVD can be more robust to computational errors compared to directly calculating eigenstates.

In this simulation, we follow the work of Yeo et al [12] and employ the SVD scheme to study the WS operator. We focus on understanding the influence of parameter X on the singular values and how they relate to the behavior of the operator.

The simulation scheme involves the following steps:

1. Measure the scattering matrix: We measure the scattering matrix of the system for a single-frequency excitation at each port.
2. Introduce variation on parameter α : A small variation on a parameter α , denoted by $\delta\alpha$, is introduced and the new scattering matrix is calculated.

3. Compute WS operator: The WS operator is computed using a finite difference method for the derivative.
4. Singular values and singular vectors Determination: Singular Value Decomposition (SVD) is employed to determine the eigenstates and eigenvalues of the operator.
5. Reconstructing the wave function: For a given eigenvalue, the corresponding eigenstate contains the coefficients for sending signals to each port. By applying an inverse Fourier transform, we determine the signal to send at each port to reconstruct the wave function associated with that eigenstate.
6. Simulate and Analyze: We run the simulation with the reconstructed wave function and calculate the average energy distribution within the system. This analysis allows us to verify if the energy distribution aligns with the expected behavior for particle-like states.

Table 3, resumes the different parameters of the simulation.

Parameter	Name	Value
f	excitation frequency	500[KHz]
λ	wavelength	0.6 [m]
Mc_s	main channel size	20λ
T_{tot}	simulation time	419.6 [ns]
dt	time step	0.1[ns]
dx	space step	$\frac{\lambda}{20}$
ϵ_r	relative permittivity	10

Table 3: Simulation Parameter Definition

2. Visualization of the eigenstates of the WS and GWS operator for our system

2.1. WS operator's eigenstates for a frequency variation

This section details our attempt to reproduce particle-like states within our 4-port system

The eigenstates of the WS operator correspond to wave functions with a well-defined dwell time within the system, as demonstrated in [10]. These eigenstates define specific propagation paths within the system, determined solely by its properties and geometry. A key advantage of our simplified 4-port system is that it facilitates the identification of these paths. Figure 4 summarizes the different trajectories.

One way to differentiate these paths effectively is by introducing the concept of apparent distance, denoted by $D_{app}(path)$. It is defined as:

$$D_{app}(path) = T(path)C_0 \quad (16)$$

Here, $T(path)$ represents the time it takes a wave to travel along the specific path. This concept is valuable because it provides a metric for comparing the path lengths. Essentially, it translates travel time into an intuitive quantity, highlighting the notion of shortest and longest propagation trajectories within the system. Table 4 summarizes the different apparent distances for each of the six identified paths as a function of the width of the slab (d) and the refractive index (η_r) normalized by the size of the main channel.

Path number	apparent distance
1	0.9
2	1
3	$1 + d (\eta_r - 1)$
4	$2.1 + d (\eta_r - 1)$
5	$1.5 + d (\eta_r - 1)$
6	1.5

Table 4: D_{app} for the different paths

Now that we have identified the paths and their corresponding apparent distances, we can verify that the eigenstates of the WS operator indeed correspond to these paths. In other words, we expect the smallest (largest) eigenvalue to correspond to the shortest (longest) apparent distance path.

While it's relatively easy to identify a correspondence between eigenstates 2 and 4 (Figures 5b and 5d, respectively) and paths number 5 and 6 (Figures 4e and 4f, respectively) Identifying a clear correspondence between eigenstates 1 and 3 (Figures 5a, 5c, respectively) and specific paths is more challenging. This discrepancy might arise because the WS operator, being of size 4x4, can only accommodate a maximum of four distinct eigenstates. However, our system has six well-defined paths. It is possible that eigenstates 1 and 3 (Figures 5a and 5c) represent a combination of the remaining paths.

This hypothesis is based on the work of Gérardin et al [10], who identified a degeneracy phenomenon in their simulations of scattering states. They employed the incident wave angle as a key parameter to differentiate between the superposed modes. This distinction was achieved by analyzing a specific subspace of the scattering matrix S .

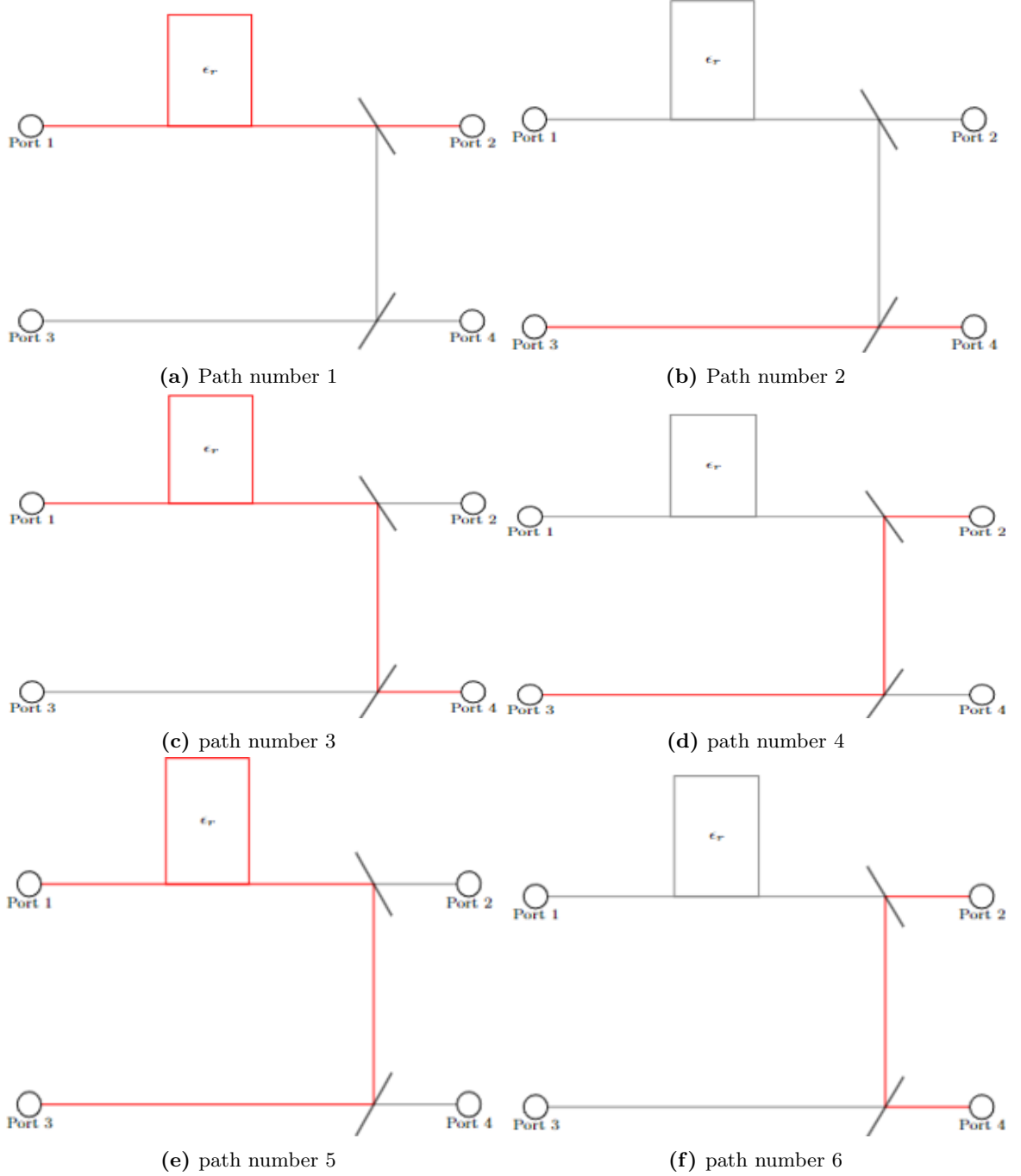


Figure 4: The six paths with well defined propagation time inside the 4-ports system

In future work, we aim to implement a similar approach to alleviate the potential degeneracy observed in our system. By adapting this technique, we hope to achieve a clearer separation of the involved states, facilitating a more comprehensive understanding of their properties. It should be noted that the eigenvalues of the WS operator may not directly correspond to the propagation times of the associated eigenstates. This discrepancy might arise because we compute the singular values of the WS operator instead of the true eigenvalues. Although Singular Value Decomposition (SVD) offers greater robustness

and applicability in various contexts, it does not necessarily guarantee an exact mapping between singular values and physical quantities such as propagation time.

2.2. GWS operator's eigenstates for variation in the slab position

In this Section, We attempt to study the generalised WS operator derived for variation in the slab's position. According to [2], the eigenstates of the GWS operator derived in this scenario corresponds to wave functions that either focus on

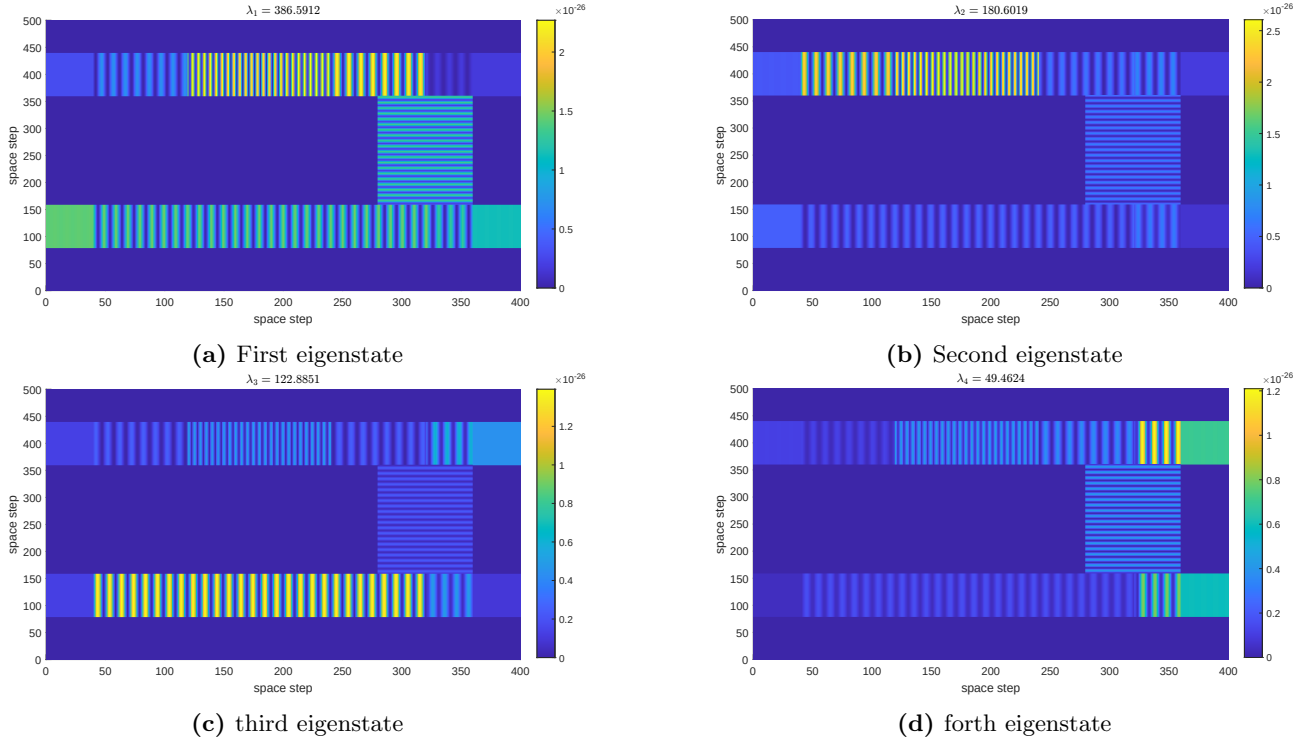


Figure 5: Average energy distribution inside the system for different modes of the WS operator for $\delta f = 1KHz$, frequency $f = 500MHz$ and $\varepsilon_r = 10$

or avoid the scatterer.

In this section, we try to reproduce this result using the same simulation scheme defined in 1.3.2 (Figure 6).

We can see that, for the first and second eigenstates (Figures 6a and 6b respectively), the corresponding wave functions focuses the energy on the slab. While for the eigenstates three and four (Figures 6c and 6d respectively), the corresponding wave functions avoid the slabs. This is in coherence with the results of Horodyski et al. [2]

Unlike in the previous section, where the classical WS operator exhibited degeneracy phenomena for frequency variation, no such phenomena were observed in the case of the GWS operator. This difference can be attributed to the size of the possibility space.

In the case of frequency variation, the size of the WS operator (4) is smaller than the size of the space of possible trajectories (6). This allows for potential overlaps or coincidences in the operator's eigenstates, leading to degeneracy (multiple states presented by the same eigenvalue).

However, for a variation on the slab position, the size of the operator (4) is larger than the size of the possibility space for scatterer position variation (2) (we either focus on or avoid the slab). In this case, the possibility space can be entirely

described by the eigenstates of the GWS operator.

In simpler terms, for the classical WS operator, the large number of possible paths compared to the size of the WS operator allows for potential overlaps, while in the case of a slab position variation, the reduced size of the possibility space allows the operator to describe each configuration without overlap.

2.3. Eigenvalue Evolution under Parameter Variations

This section focuses on the behavior of eigenvalues as functions of two types of variations: changes in wave frequency (δf) and variations in slab position (δp).

Although the 1D (FDTD) code with wave dividers provided a simplified framework for understanding the behavior of the WS operator, it has limitations in fully capturing the true physics of the system. These limitations can potentially influence the predicted evolution of the system's eigenstates.

To address this issue and obtain more accurate results, we utilize a 2D monomode simulation in Comsol for the same system presented in Figure 3. This 2D simulation offers a more realistic representation of the system, leading to a more reliable prediction of the eigenvalue evolu-

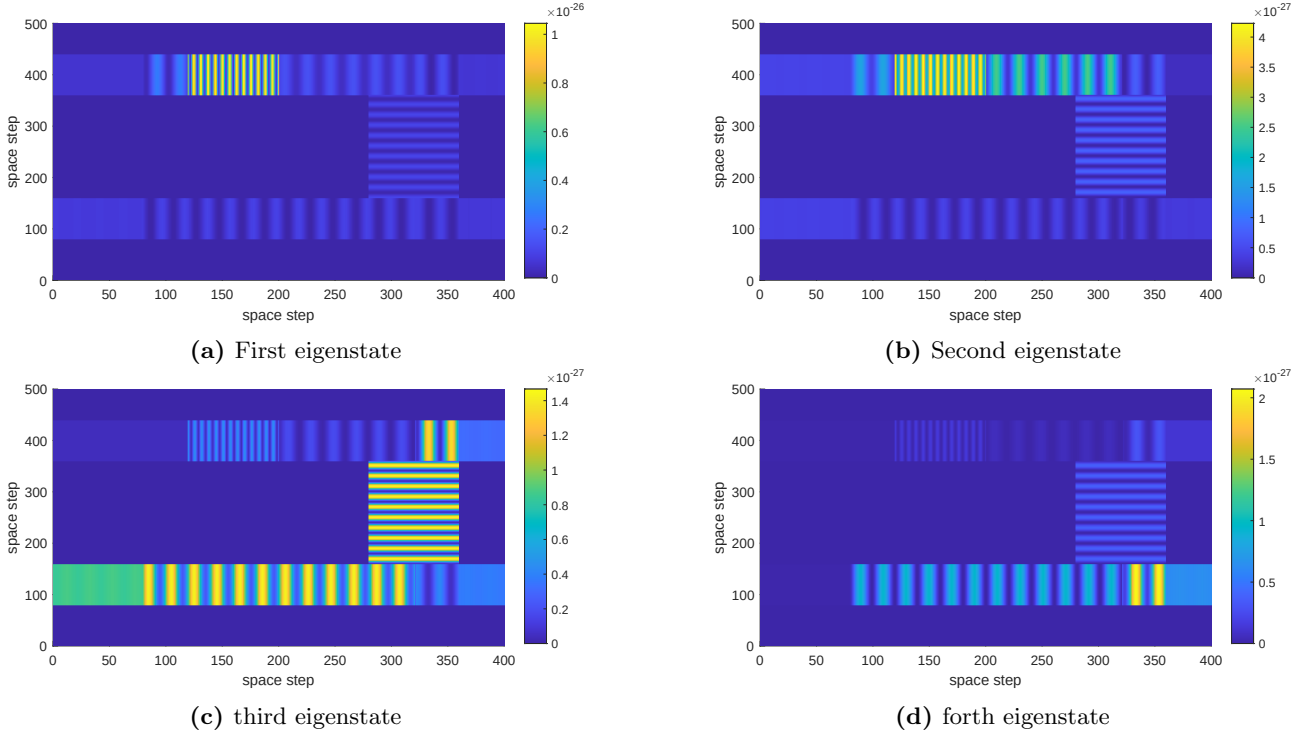


Figure 6: Average energy distribution inside the system for different modes of the GWS derived for a variation of the slab position of $\Delta p = \frac{\lambda}{100}$ for a frequency $f = 500MHz$ and relative permittivity $\epsilon_r = 10$.

tion under parameter variations.

2.3.1. WS operator's eigenvalues evolution for frequency variation

As discussed earlier, the eigenstates of (WS) operator correspond to particle-like states with well-defined dwell times within the system [10]. These propagation modes, determined solely by the system's properties and not by the specific wave characteristics, define the paths taken by the waves. Consequently, as long as the linear approximation remains valid, the eigenvalues of the WS operator are expected to be independent of the amplitude of the introduced variations δf . This prediction is confirmed with the simulation (Figure 7).

2.3.2. GWs operator's eigenvalues evolution for slab's position variation

While the eigenvalues of the Wigner-Smith (WS) operator characterize intrinsic properties of the system, the eigenvalues of the Generalized Wigner-Smith (GWS) operator, particularly Q_p , computed for variations in slab position, reflect the intensity of the force exerted by the wave on the scatterer. As previously discussed, the first two eigenstates associated with the highest eigenvalues of Q_p correspond to particle-like wave

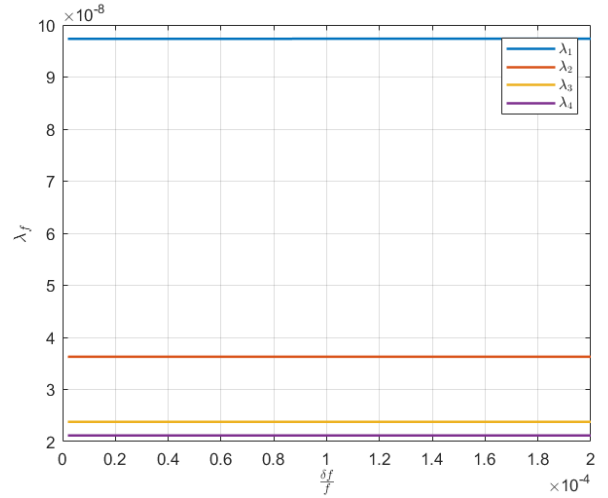


Figure 7: WS eigenvalues evolution for a main frequency $f = 500MHz$ and a frequency variation δf ranging from $0.01MHz$ and $0.1MHz$

functions that concentrate on the slab, exerting the strongest force. Intuitively, one might expect a direct proportionality between the variation amplitude and the eigenvalue magnitude.

However, an unexpected behavior emerges when analyzing the evolution of the eigenvalues for different variations in slab position (Figure 8):

1. Eigenvalues 3 and 4 maintain very small values, consistent with particle-like states that

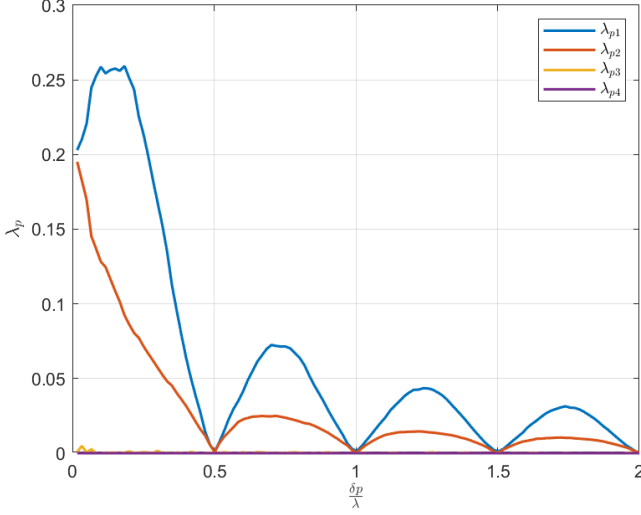


Figure 8: Q_p eigenvalues evolution as a function of the δp varying across a range of 2λ for frequency $f = 500\text{MHz}$ and $\varepsilon_r = 10$.

avoid the slab.

2. Eigenvalues 1 and 2, representing states focused on the slab and exerting the maximum force, exhibit a pseudo-periodic behavior with a period of $\frac{\lambda}{2}$ (half the wavelength).

This behavior can be partially understood if we interpret the variation of the slab position as the appearance and disappearance of 2 slabs both of width δp . Figure 9 illustrate this concept.

Considering that the eigenstates associated with the highest eigenvalues of the GWS operator (Q_p) correspond to wave functions that concentrate on the newly created slab (represented by the blue region in figure 9), the observed $\lambda/2$ periodicity in these eigenvalues becomes more understandable.

The force exerted by a wave on the slab originates from the momentum change experienced by the wave particles during reflection at the slab's interfaces. Consequently, if the reflection coefficient of the slab were zero, the wave would not exert any force. In such a scenario, one might expect the corresponding eigenvalues (which reflect this force intensity) to be null as well.

Interestingly, the reflection coefficient of a slab can approach zero when its width is an integer multiple of $\lambda/2$. This observation aligns with the observed periodicity $\lambda/2$ in the eigenvalues associated with the strongest force interaction [11].

The previous interpretation provides a compelling explanation for the observed periodicity $\lambda/2$ in the eigenvalues associated with the strongest force interaction. However, it does not fully explain the observed variations in force intensity within each period.

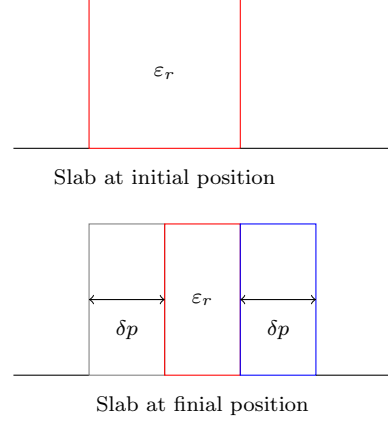


Figure 9: Slab Position variation : the displacement of the slab could be interpreted as the creation of a new slab (Blue) and the disappearance of a slab (Gray) both of width δp .

This discrepancy suggests that additional mechanisms might be at work. One potential explanation is the introduction of dispersion effects in 2D simulations. Dispersion refers to the phenomenon in which the propagation characteristics of a wave depend on its frequency. This dependence can potentially modulate the force exerted by the wave on the slab, resulting in the observed variations within each $\lambda/2$ period.

3. Sensitivity Analysis of the Combined Effect of Frequency and Slab position variations on the GWS Operator

3.1. Qualitative Analysis of the WS operator Sensitivity

The most common approach for computing the WS operator derivative is based on the finite difference method applied to the scattering matrix. Experimentally, this involves evaluating the scattering matrix in two different configurations: $\alpha = \alpha_1$ and $\alpha = \alpha_1 + \delta\alpha$, where α represents the variable of interest and $\delta\alpha$ is a small variation. The derivative of the scattering matrix with respect to α is then approximated using the following equation:

$$\frac{\partial S}{\partial \alpha} \approx \frac{S(\alpha_1 + \delta\alpha) - S(\alpha_1)}{\delta\alpha} \quad (17)$$

While this method offers a straightforward computational scheme, it possesses certain limitations:

1. The underlying approximation used to calculate the derivative becomes increasingly inaccurate for larger variations of $\delta\alpha$

2. If the variation $\delta\alpha$ necessitates interacting with the system, it can inadvertently alter other global parameters β beyond the intended variable. These unintended modifications can introduce unwanted artifacts into the results.

In this section we focus on the second point, we try to characterize the effect of introducing a secondary change.

A qualitative analysis of this scenario suggests that the impact of the unintended variation ($\delta\beta$) on the behavior of the system depends on its amplitude:

1. **Small Amplitude ($\delta\beta$):** When the amplitude of $\delta\beta$ is small, it might not significantly influence the eigenvalues of the pure GWS operator (Q_α). In this case, we expect that the eigenstates associated with Q_α remain relatively unchanged, resulting in an error term that can potentially lead to inaccurate predictions about the system's behavior.
2. **Large Amplitude ($\delta\beta$):** If the amplitude of $\delta\beta$ is sufficiently large, the unintended variation can significantly alter the system itself. This significant change can manifest as a modification of the system's behavior and a change in the eigenvalues of Q_α . Consequently, we expect that the eigenstates associated with the GWS operator will also be altered.

To illustrate this qualitative prediction further, consider the system introduced in Section 3. Table 4 summarizes the apparent distances for the six different propagation modes. We will explore the impact of an unintended variation in slab width (δd) in a system where the main variable of interest is the variation in wave frequency (δf).

In this scenario, the eigenstates of the WS operator with respect to frequency (Q_f) exhibit particle-like behavior, primarily favoring the shortest or longest paths. However, it is important to note that the apparent lengths of these paths themselves depend on the actual width of the slab. Therefore, if the variation in slab width (δd) is small, it could not alter the order of the preferred paths. Instead, it would likely introduce an error term in the eigenstates of Q_f . However, if the magnitude of δd exceeds a certain threshold, this unintended variation can significantly modify the system. This significant change can manifest itself as a complete reversal in the order of preferred paths, leading to a substantial alteration of the eigenstates associated with Q_f . Table 5 illustrates this change; here we

choose $\varepsilon_r = 9$ for the relative permittivity of the slab.

	$\delta d = 0.025$		$\delta d = 0.1$	
Path number	D_{app}	Order	D_{app}	Order
1	0.90	1	0.9	1
2	1.00	2	1.0	2
3	1.45	3	1.6	4
4	2.55	6	2.7	6
5	1.95	5	2.1	5
6	1.50	4	1.5	3

Table 5: Apparent distances (D_{app}) and path orders for different variations in slab width (δd). This table illustrates how variations in slab width can affect propagation modes within the system. For small variations (e.g., $\delta d = 0.025$), the path orders likely remain unchanged, resulting in minimal influence on the Wigner-Smith (WS) operator's eigenstates. However, larger variations (e.g. $\delta d = 0.1$) can modify the path orders, potentially leading to significant changes in the WS operator's eigenstates and eigenvalues.

3.2. Analytical characterization of multi variable change on GWS operator

Consider a scenario where an undesired change is introduced to variable β while calculating the GWS operator for variable α . If not taken into account, one may fall into the following error while calculating the derivative 18.

$$\frac{\partial S}{\partial \alpha} \approx \frac{S(\alpha + \delta\alpha, \beta + \delta\beta) - S(\alpha, \beta)}{\delta\alpha} \quad (18)$$

Here, $\delta\alpha$ and $\delta\beta$ represent small variations in the respective variables. Equation 18 highlights the approximation involved in the finite difference method when unintended parameter variations occur.

Based on this approximation, the incorrectly calculated GWS operator can be expressed as (see the Appendix 4.2 for more details):

$$Q \approx Q_\alpha + Q_\beta \frac{\delta\beta}{\delta\alpha} + Q_{\alpha,\beta} \delta\beta \quad (19)$$

where:

1. Q_α represents the true GWS operator with respect to variable α .
2. Q_β represents the contribution due to the unintended variation in β .
3. $Q_{\alpha,\beta} = -iS^\dagger \frac{\partial^2 S}{\partial \alpha \partial \beta}$ captures the coupling between the two variables α and β .

a striking result is that the calculated operator Q is not ,generally , Hermitien. This is due to the presence of the coupling operator $Q_{\alpha,\beta}$. In fact, one can only preserve this property, essential for the interpretation of the eigenvalues of the operator, only in two cases

3.2.1. First case $Q_{\alpha,\beta} = 0$

The first case, $Q_{\alpha,\beta} = 0$, could be written as

$$\frac{\partial^2 S}{\partial \alpha \partial \beta} = 0 \quad (20)$$

This implies that the scattering matrix S can be written as a linear combination

$$S = AS_1(\alpha) + BS_2(\beta) \quad (21)$$

a system described by equation 21, can be represented by the following figure 10.

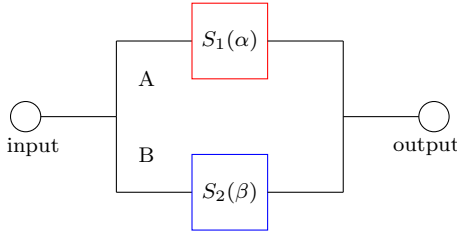


Figure 10: Basic design for a scattering system described by equation 21 : The system has a parallel design with two independent scattering regions, resulting in a null coupling operator

3.2.2. Second case

The second case (equation 22) for which the operator Q is also hermitian can be derived from equation 3 (See appendix 4.2 for more details).

$$\frac{\partial S^\dagger}{\partial \alpha} \frac{\partial S}{\partial \beta} = -\frac{\partial S^\dagger}{\partial \beta} \frac{\partial S}{\partial \alpha} \quad (22)$$

This condition can also be written as

$$A_{\alpha,\beta}^\dagger = A_{\alpha,\beta} \quad (23)$$

where

$$A_{\alpha,\beta} = -i \frac{\partial S^\dagger}{\partial \alpha} \frac{\partial S}{\partial \beta} \quad (24)$$

While in the first case a physical interpretation was straightforward, in the second case it is harder to imagine what kind a system satisfies these equations. More work is needed to further understand the theoretical significance of this new operator and its proprieties.

The next section will focus on validating these analytical predictions using Comsol simulations. We aim to characterize the impact of introducing a secondary variation on the eigenstates and eigenvalues of WS operator, derived for a change in wave frequency . We focus on scenarios where the introduced variations in the position of the slab (δp) are small enough that the order of the propagation modes (paths) remains unchanged.

3.3. Simulation analysis of multi variable variation on GWS operator

Following the development of a mathematical framework in the previous section to analyze the impact of multi-variable changes on the GWS operator behavior, this section focuses on the influence of unwanted variation on the eigenvalues of the GWS operator.

According to equation 19 the error introduced by Q_β is modulated by $\frac{\delta \beta}{\delta \alpha}$. This implies that the dominant influence is not directly proportional to the absolute value of $\delta \beta$, but rather its relative importance compared to $\delta \alpha$. In other words, if one is primarily interested in relatively large variations of the main variable ($\delta \alpha$), one can initially neglect the influence of secondary variables for first-order accuracy.

Figures 11 and 12 further substantiate our observations. Focusing on the evolution of the first few eigenvalues:

1. λ_{fp1} (Figure 12a) with a variation $\delta f = 1$ kHz exhibits a larger deviation amplitude from the original first eigenvalue of Q_f , λ_{f1} , compared to the case with $\delta f = 1$ MHz (Figure 11a).
2. This trend holds for eigenvalues 2 and 3 as well, where the deviations are more pronounced in the 1 kHz case.
3. Eigenvalue 4, however, appears to remain closer to its original value, λ_{f4} , in both scenarios, exhibiting only minor periodic fluctuations.

These observations align with our theoretical prediction that the influence of unintended position variation (δp) is modulated by the relative amplitude $\frac{\delta p}{\delta f}$. A larger frequency variation (smaller relative amplitude) leads to a weaker influence of the position variation on the eigenvalues, resulting in less pronounced deviations.

Figure 12 reveals another key observation: the unintended position variation (δp) does not affect all eigenvalues in the same way. The periodic pattern for the first eigenvalue, λ_{fp1} , appears relatively smooth. However, for eigenvalues 2 and

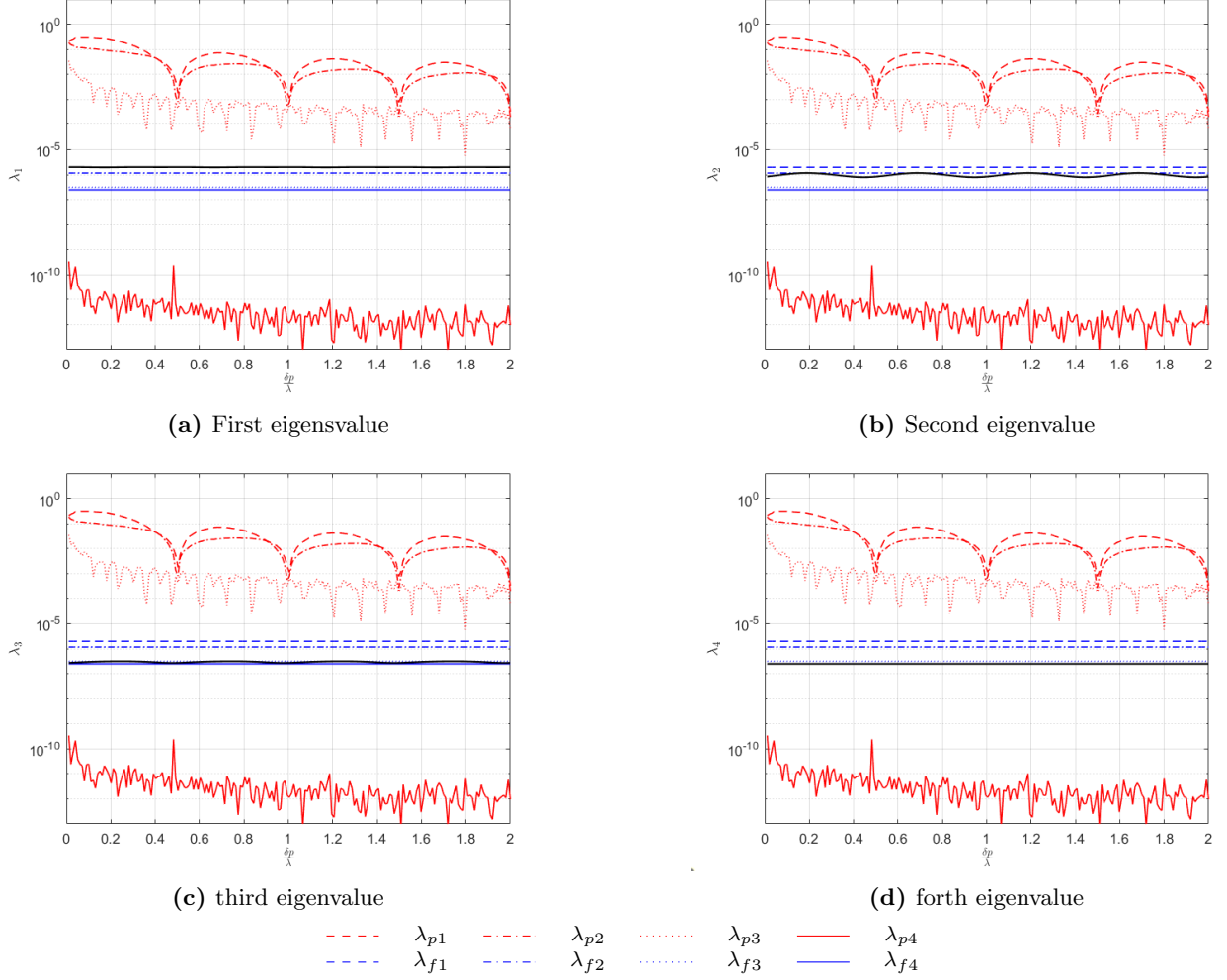


Figure 11: GWS eigenvalue evolution with frequency and position variation for a frequency variation of $\delta f = 1$ MHz and a position variation δp across a range of 2λ . The eigenvalues are plotted on a semi-logarithmic axis with a main frequency $f = 500\text{MHz}$ and slab relative permittivity $\varepsilon_r = 25$.

3 specifically, a more pronounced flattening of the periodic behavior is evident. This suggests a potential dependence of the influence from δp on the specific eigenvalue itself, beyond the simple relationship with the relative amplitude, $\frac{\delta\beta}{\delta\alpha}$. Further investigation into this behavior might be necessary.

a final observation from Figure 12 indicates that although the eigenvalues λ_{fp1} , λ_{fp2} , λ_{fp3} have the same period as the eigenvalues λ_{p1} and λ_{p2} , a phase change appears to be introduced.

More work is needed to further understand how the eigenvalues of Q_f and Q_p influence each other.

4. Summary and Futur Work

4.1. Summary

This study explored the Wigner-Smith (WS) operator and its application in the electromagnetic domain to identify open and closed propagation channels for a scattering system. We began with a two-port system before introducing a four-port system. This allowed for a deeper investigation of particle-like states for both the classical WS operator derived for a frequency variation and the Generalized Wigner-Smith (GWS) operator derived for a variation in slab position.

In both cases, we have analyzed how the eigenvalues of the operator evolve with the introduced variation.

Our findings revealed that the eigenvalues of the classical WS operator remained unchanged. This aligns with theoretical predictions as they represent a system property (path propagation

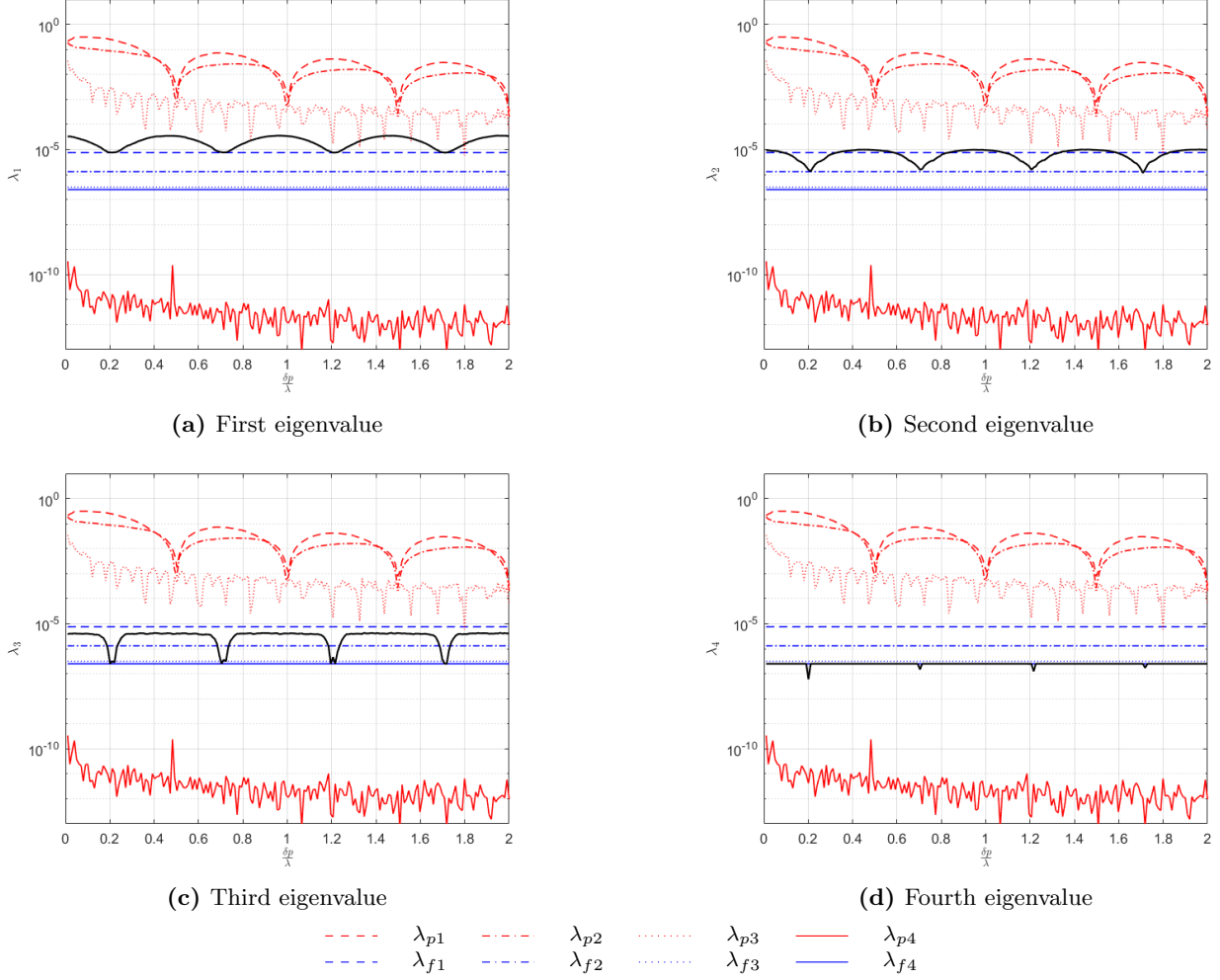


Figure 12: GWS Eigenvalues Evolution at a frequency variation of $\delta f = 1$ kHz and a position variation δp across a range of 2λ . The eigenvalues are plotted on a semi-logarithmic axis with a main frequency $f = 500\text{MHz}$ and slab relative permittivity $\varepsilon_r = 25$.

time) independent of the wave itself 7.

For the GWS operator derived with a slab position variation, we observed the following:

1. The two lowest eigenvalues (associated with particle-like states avoiding the slab) remained unchanged.
2. The two largest eigenvalues (associated with particle-like states focusing on the slab) exhibited a pseudo-periodic behavior. This behavior is partially explained by the properties of the slab reflection coefficient 8.

The final section of this work focused on the GWS operator's sensitivity to unintended changes. We partially developed a mathematical framework and identified two key findings.

1. **Loss of Hermiticity:** The GWS operator, derived for multivariable changes, generally loses its fundamental property of hermiticity.

Two cases were identified in which this property is conserved:

- (a) When the scattering system can be described by two independent subsystems in a parallel configuration, ensuring a null coupling operator.
- (b) When the scattering matrix satisfies a specific condition (further details on these cases will be provided in a future publication).

2. **Relative Amplitude Modulation:** The influence of the unintended variation is modulated by the relative amplitude of the variations ($\frac{\delta\beta}{\delta\alpha}$). This observation was validated using Comsol simulations.

These findings offer valuable insights into the behavior and sensitivity of the GWS operator in scattering systems.

4.2. Future Work

Although this study offers valuable insights into the GWS operator, several intriguing questions remain open, paving the way for future research directions:

- 1. Degeneracy in Eigenstates of WS Operator:** A deeper investigation is needed into the degeneracy phenomena observed in the eigenstates of the WS operator when the operator size is smaller than the number of paths with well-defined dwell times. Understanding this behavior will provide a more comprehensive picture of the WS operator's properties.
- 2. Scatterer Movement and Microobject Manipulation:** Studying the evolution of eigenvalues when a scatterer is moved within the system can offer valuable insights into the manipulation of micro objects. This knowledge could be instrumental in developing novel techniques for microfluidic and nanophotonic applications.
- 3. GWS Operator with Multi-Variable Changes:** A more comprehensive theoretical understanding of the GWS operator under multi-variable changes is crucial. This includes:
 - (a) Hermiticity Preservation: A deeper interpretation of the conditions that allow the operator to remain Hermitian is required, along with their physical implications. Validation of these conditions through simulations would solidify the theoretical framework.
 - (b) Eigenvalue Interaction: Exploring how the different eigenvalues influence each other for multi-variable changes is essential. Furthermore, understanding the physical interpretation of the WS operator eigenvalues in this context and the possibility of predicting them would be highly valuable.
- 4. Eigenvector Evolution and Wave Shape Control:** A theoretical and simulation-based study of the GWS operator's eigenvector evolution is warranted. Since eigenvectors control wave shape, a deeper understanding of their behavior will provide valuable insights into wave manipulation using the GWS operator.
- 5. Influence of port number on eigenstates generation:** Understanding how the number of ports and consequently the size of the GWS

operator can influence the determination of GWS operator eigenstates is essential from both theoretical and experimental perspectives. Analytically determining the minimum number of ports required to generate specific waveshapes can significantly enhance the controllability of experimental setups.

- 6. GWS Operator for Multi-Feature Waveform Design:** This work presented the GWS multi-variable study as a sensitivity analysis. However, one could envision designing a waveform with two distinct features, such as crossing the system quickly while focusing on a target area. The mathematical tools presented here could serve as a starting point for investigating the feasibility of achieving such complex wave behaviors using multi-variable GWS operator control.

The GWS operator has immense potential for various applications. As this study demonstrates, numerous unanswered questions require further exploration. Future research efforts, both theoretical and experimental, along with numerical simulations, are crucial to unlocking the full potential of the GWS operator and its capabilities in wave manipulation.

Appendix

Appendix A : Particle-Like State Approach to WS Operator Derivation

In this section, we present a simplified derivation for the classical WS operator. As established in Section 2.3.1, the eigenvalues of the WS operator exhibit first-order independence from frequency variations. Leveraging the properties of particle-like wave states, this independence property can potentially be extended to the eigenvectors as well.

Particle-like wave states, characterized by their trajectory within the scattering system, are independent of frequency. However, their global properties, overall phase and amplitude, can exhibit changes. Mathematically, for an wave vector denoted by Φ_{in} , the associated output wave vector is particle like state if it verefy the propriety expressed by:

$$\Phi_{out}(f + \delta f) = z\Phi_{out}(f) \quad (25)$$

where z is a complex number. applying a first order Taylor development to equation 25 leads to

$$\Phi_{out}(f) + \frac{\partial \Phi_{out}}{\partial f} \delta f = z\Phi_{out} \quad (26)$$

This can also be written as :

$$\frac{\partial \Phi_{out}}{\partial f} \delta f = z' \Phi_{out} \quad (27)$$

Using equation 1 this condition can be written in terms of the scattering matrix S :

$$\frac{\partial S}{\partial f} \Phi_{in} \delta f = z' S \Phi_{in} \quad (28)$$

Here we assume that the input vector is independent of the frequency. Using equation 3, we can finally write :

$$-iS^\dagger \frac{\partial S}{\partial f} \Phi_{in} = \frac{z'}{i\delta f} \Phi_{in} \quad (29)$$

Hence, the input wave vector that exhibits particle-like states is a eigenvector of the WS operator.

Appendix B : Derivation of equation 19

In this section we go through the steps necessary to derive equation 19 from equation 18. As discussed earlier, equation 18 is an inaccurate calculation of the scattering matrix derivative if the undesired variation $\delta\beta$ is not taken into account. In this case using a first order Taylor development with respect to $\delta\alpha$ and $\delta\beta$ leads to

$$S(\alpha + \delta\alpha, \beta + \delta\beta) \approx S(\alpha, \beta) + \frac{\partial S}{\partial \alpha} \delta\alpha + \frac{\partial S}{\partial \beta} \delta\beta + \frac{\partial^2 S}{\partial \alpha \partial \beta} \delta\alpha \delta\beta \quad (30)$$

If we considered this development for calculating the derivative, then when applying the definition of the GWS operator derived essentially for a variation of α then we find that :

$$\begin{aligned} & -iS^\dagger \frac{S(\alpha + \delta\alpha, \beta + \delta\beta) - S(\alpha, \beta)}{\delta\alpha} \approx \\ & -iS^\dagger \frac{\partial S}{\partial \alpha} + -iS^\dagger \frac{\partial S}{\partial \beta} \delta\beta + -iS^\dagger \frac{\partial^2 S}{\partial \alpha \partial \beta} \delta\beta \end{aligned} \quad (31)$$

Interpreting this equation in terms of GWS operator we find equation 19

$$Q \approx Q_\alpha + Q_\beta \frac{\delta\beta}{\delta\alpha} + Q_{\alpha,\beta} \delta\beta \quad (19)$$

Appendix C : Proof of Hermiticity for the Second Case

In this section we proof that the coupling operator $Q_{\alpha,\beta}$ is Hermitian when the scattering matrix

verify the condition 22. For this we start from equation 3.

$$S^\dagger S = I \quad (3)$$

Taking the derivative of this equation with respect to α we find

$$\frac{\partial S^\dagger}{\partial \alpha} S + S^\dagger \frac{\partial S}{\partial \alpha} = 0 \quad (32)$$

If we take the derivative of this equation with respect to β we find :

$$\frac{\partial^2 S^\dagger}{\partial \alpha \partial \beta} S + \frac{S^\dagger}{\partial \alpha} \frac{\partial S}{\partial \beta} + \frac{S^\dagger}{\partial \beta} \frac{\partial S}{\partial \alpha} + S^\dagger \frac{\partial^2 S}{\partial \alpha \partial \beta} = 0 \quad (33)$$

multiplying this equation by i we can write it in terms of the coupling operator :

$$Q_{\alpha,\beta}^\dagger - Q_{\alpha,\beta} + A_{\alpha,\beta}^\dagger - A_{\alpha,\beta} = 0 \quad (34)$$

Where A is the operator defined in equation 2. Equation 34 make it clear that $Q_{\alpha,\beta}$ is only hermitian only if A is also hermitian.

Supplementary materials

A first differential equation for the evolution of eigenstates under change in multiple variables

Here we try to develop a system of equations that could describe how the eigenvector (q_α) and the eigenvalue (λ_α) evolve with a change in the parameter β . we start from the fundamental equation linking q_α and λ_α

$$Q_\alpha q_\alpha = \lambda_\alpha q_\alpha \quad (35)$$

Deriving this equation with respect to β

$$\frac{\partial(Q_\alpha q_\alpha)}{\partial \beta} = \frac{\partial(\lambda_\alpha q_\alpha)}{\partial \beta} \quad (36)$$

$$\frac{\partial Q_\alpha}{\partial \beta} q_\alpha + Q_\alpha \frac{\partial q_\alpha}{\partial \beta} = \frac{\partial \lambda_\alpha}{\partial \beta} q_\alpha + \lambda_\alpha \frac{\partial q_\alpha}{\partial \beta} \quad (37)$$

We define the $M_{\alpha,\beta}$ operator as

$$M_{\alpha,\beta} = \frac{\partial Q_\alpha}{\partial \beta} = A_{\alpha,\beta} + Q_{\alpha,\beta} \quad (38)$$

Finally we can write a first differential equation linking the evolution of q_α and λ_α .

$$(M_{\alpha,\beta} - \frac{\partial \lambda_\alpha}{\partial \beta} I) q_\alpha = (\lambda_\alpha I - Q_\alpha) \frac{\partial q_\alpha}{\partial \beta}. \quad (39)$$

A second equation is needed to describe how λ_α evolves in order to solve for the evolution of eigenvalues and states.

Insights from Quantum physics on Multivariate Variations

An important insight into the Hermiticity property of the coupling operator $Q_{\alpha,\beta}$ comes from quantum mechanics. In quantum physics, the Hermiticity of an operator implies that its eigenvalues correspond to physical properties that are measurable experimentally. Therefore, in cases where the coupling operator is Hermitian, its eigenvalues could represent physical properties of the system. It remains unclear whether these eigenstates or eigenvalues are related to the eigenstates and eigenvalues of Q_α and Q_β or if they represent different properties of the system under investigation.

This could become clear if one can find a system design for which the second Hermitian condition is verified [22](#).

A first interesting insight comes from a deeper investigation of the operator $A_{\alpha,\beta}$. In fact, one can link $A_{\alpha,\beta}$ to the individual GWS operators Q_α and Q_β as follows:

$$Q_\alpha Q_\beta = -iA_{\alpha,\beta} \quad (40)$$

Hence the herminity condition is a equivalent to anticommutation of the the individual operator Q_α and Q_β ie $Q_{\alpha,\beta}$ is hermitian is equivalent to

$$\{Q_\alpha, Q_\beta\} = Q_\alpha Q_\beta + Q_\beta Q_\alpha = 0 \quad (41)$$

In quantum physics, the anticommutation of two operators has significant consequences: If we consider an eigenstate q_α of the operator Q_α associated with an eigenvalue λ_α , then using the anti-commutation property we can deduce that $Q_\beta q_\alpha$ is another eigenstate of Q_α associated with the eigenvalue $-\lambda_\alpha$. Using this, we can deduce that no GWS operator could anticommute with Q_f : Given that the eigenvalues of Q_f represent the travel time of an eigenstate q_f , if a GWS operator Q_β anticommutes with Q_f , the new eigenstates will have a negative propagation time, which is absurd. Hence, no GWS operator could verify the anticommutation property with Q_f .

a Potential system design that verifies condition [22](#)

In this Section we explore a potential system design that could verify the anti commutation condition

$$\frac{\partial S^\dagger}{\partial \alpha} \frac{\partial S}{\partial \beta} = -\frac{\partial S^\dagger}{\partial \beta} \frac{\partial S}{\partial \alpha} \quad (22)$$

Instead of looking for a general solution of the such equation we can note that a scattering ma-

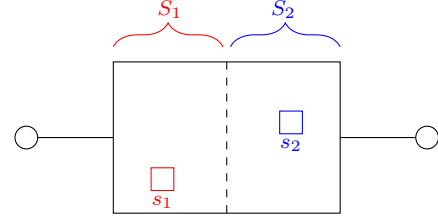


Figure 13: Possible system design that verifies the anti-commutation condition

trix S verifies this conditions if

$$\frac{\partial S}{\partial \alpha} = i \frac{\partial S}{\partial \beta} \quad (42)$$

In order to go further one can look for solution where the matrix S takes the form of :

$$S(\alpha, \beta) = S_1(\alpha)S_2(\beta) \quad (43)$$

To get better idea of what systems have matrix S that could be described by equation [43](#), Imagine a system [13](#) with two scatters s_1 and s_2 . The scattering matrix S depends on the positions of both of the them α and β . Clearly for this geometry to be valid the matrices $S_1(\alpha)$ and $S_2(\beta)$ should commute.

Inserting this new form into equation [42](#) implies that :

$$S_1^\dagger \frac{\partial S_1}{\partial \alpha} = i \frac{\partial S_2}{\partial \beta} S_2^\dagger \quad (44)$$

the left side only depends on α while the right side only depends on β this can only be verified if both sides are equal to a constant matrix B . Hence we end up with 2 differential equations :

$$\frac{\partial S_1}{\partial \alpha} = S_1 B \quad (45)$$

$$\frac{\partial S_2}{\partial \beta} = -i B S_2 \quad (46)$$

And finally we get :

$$S_1(\alpha) = S_1(0)e^{B\alpha} \quad (47)$$

$$S_2(\beta) = e^{-iB\beta} S_2(0) \quad (48)$$

In order for this development to work B should be a hermitian matrix. Now that we have an idea of a possible system that verify the anti-commutation propriety we can go more in details of the study of the coupling operator.

References

- [1] John Pendry. Light finds a way through the maze. *Physics*, 1, 09 2008.
- [2] Michael Horodyski, Matthias Kühmayer, Andre Brandstötter, Kevin Pichler, Yan V. Fyodorov, Ulrich Kuhl, and Stefan Rotter. Optimal wave fields for micromanipulation in complex scattering environments. *Nat. Photonics*, 14(3):149–153, March 2020.
- [3] Harold U. Baranger and Pier A. Mello. Mesoscopic transport through chaotic cavities: A random s-matrix theory approach. *Phys. Rev. Lett.*, 73:142–145, Jul 1994.
- [4] Shoji Maruo, Koji Ikuta, and Hayato Korigi. Submicron manipulation tools driven by light in a liquid. *Applied Physics Letters*, 82(1):133–135, 01 2003.
- [5] Philipp Ambichl, Andre Brandstötter, Julian Böhm, Matthias Kühmayer, Ulrich Kuhl, and Stefan Rotter. Focusing inside disordered media with the generalized wigner-smith operator. *Phys. Rev. Lett.*, 119:033903, Jul 2017.
- [6] Eugene P. Wigner. Lower limit for the energy derivative of the scattering phase shift. *Phys. Rev.*, 98:145–147, Apr 1955.
- [7] Felix T. Smith. Lifetime matrix in collision theory. *Phys. Rev.*, 118:349–356, Apr 1960.
- [8] Utkarsh R. Patel and Eric Michielssen. Wigner–smith time-delay matrix for electromagnetics: Theory and phenomenology. *IEEE Transactions on Antennas and Propagation*, 69(2):902–917, 2021.
- [9] Stefan Rotter, Philipp Ambichl, and Florian Libisch. Generating particlelike scattering states in wave transport. *Phys. Rev. Lett.*, 106:120602, Mar 2011.
- [10] Benoît Gérardin, Jérôme Laurent, Philipp Ambichl, Claire Prada, Stefan Rotter, and Alexandre Aubry. Particlelike wave packets in complex scattering systems. *Phys. Rev. B*, 94:014209, Jul 2016.
- [11] S Orfanidis. Electromagnetic waves and antennas (sophocles j. orfanidis, 2016). *URL <https://books.google.com.au/books>*, 2018.
- [12] K. Brahima Yeo, Cecile Leconte, Philipp del Hougne, Philippe Besnier, and Matthieu Davy. Time reversal communications with channel state information estimated from impedance modulation at the receiver. *IEEE Access*, 10:91119–91126, 2022.